

High Availability (HA)

High Availability (HA) refers to the capability of a system or infrastructure to remain continuously operational and accessible, even during failures or under heavy load conditions. A high-availability infrastructure is not only designed to resist persistent cyberattacks but is also equipped with the intelligence to detect, mitigate, and self-heal vulnerabilities in real-time. This ensures uninterrupted services, robust data security, and the integrity of critical operations despite evolving cybersecurity threats.

One of the core components supporting high availability in web infrastructure is the **network load balancer**. A load balancer is responsible for distributing a large volume of incoming web traffic efficiently across multiple backend servers, ensuring that no single server becomes a bottleneck or point of failure.

Network Load Balancer

As shown in Figure 13.1, a **load balancer** receives incoming traffic via a virtual IP (VIP) and intelligently routes requests to one of the available web servers (Web 1 to Web 4). It ensures:

- **Optimal resource usage** by forwarding requests to the **least utilized host** (e.g., Web 3 in the diagram with only 50 current requests).
- **Server availability** by avoiding servers that are offline (e.g., Web 4).
- **Fast response time** by reducing load on any single server, allowing web pages to load rapidly for users.
- **Seamless user experience** by possibly directing different user requests to different servers, improving performance and scalability.

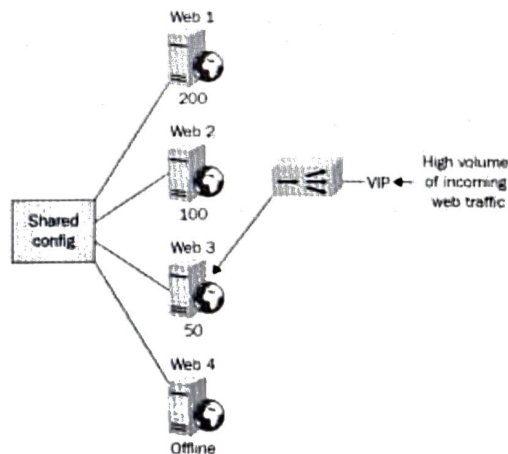


Figure 13.1: Load balancer layout

Load Balancing Techniques

1. **Least Utilized Host:** The load balancer monitors server workloads and forwards requests to the server with the lightest load. This method adapts to changing server utilization patterns and is effective for dynamic environments.

2. **Affinity (Session Persistence):** Also known as sticky sessions, this technique ensures that a client's subsequent requests during a session are directed to the same server, identified using the client's IP address or session token. This is useful for applications that maintain session state.
3. **DNS Round Robin (Figure 13.2):** In this method, the Domain Name System (DNS) provides multiple IP addresses (e.g., for Web 1, Web 2, Web 3) in a rotating sequence. Each incoming request is resolved to a different IP address in turn, distributing the load across servers. As shown in Figure 13.2, this helps balance the load at the DNS resolution level before the load balancer handles the traffic.

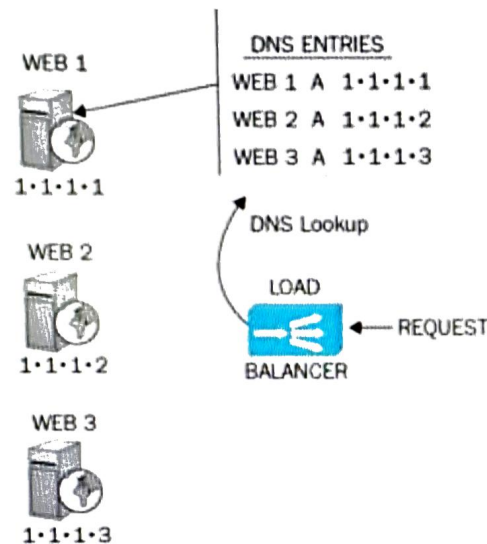


Figure 13.2: DNS round robin

Conclusion

High availability is crucial in today's digitally connected world. With techniques like **load balancing**, **affinity routing**, and **DNS round robin**, systems can maintain service continuity, prevent downtime, and deliver reliable performance—even under intense traffic or attack. These strategies collectively enhance the resilience, efficiency, and security of web services.